

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0443

Barling 345 kV
October 02, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0443. The Interconnection Customer proposes a 147.5 MW solar photovoltaic facility interconnecting at the Barling 345 kV substation in the Maryland area. The request was received on March 10, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 223 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,670,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ash Street Station 223 – Tombstone 345.0 kV	Tombstone – Marcus Hook Energy Lp 500.0 kV line	105.2	1,197	6.9
Tombstone – Marcus Hook Energy Lp 500.0 kV	Connell – Tombstone 345.0 kV Line	104.2	2,839	10.2
Sharyland – Bixb Hill 345.0 kV	Bixby Hill – Tombstone 500.0 kV	103.8	2,557	13.2
Tombstone – Sharyland 345.0 kV	Bixby Hill – Tombstone 500.0 kV	103.1	2,612	12.3
Irvie – Chiquita 345.0 kV	Marcus Hook Energy Lp – Tombstone 500.0 kV	100.1	1,462	5.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ash Street Station 223 – Tombstone 345.0 kV	Reconductor line	\$10,236,000	2% (\$255,000)
Marcus Hook Energy Lp Substation – Tombstone 500.0 kV line	Reconductor line	\$26,287,000	1% (\$289,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sharyland – Bixb Hill 345.0 kV	Reconductor line	\$14,392,000	2% (\$216,000)
Tombstone – Sharyland 345.0 kV	Reconductor line	\$14,113,000	1% (\$191,000)
Irvine – Chiquita 345.0 kV	Reconductor line	\$33,864,000	2% (\$719,000)
	Total allocated to project		\$1,670,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.